

Point of view: No significant changes in LED testing hardware

LED testing hardware has not changed much in recent years. However, the changing demand within the industry has helped companies innovate and improve their software. All products that we sell are used within our own testing laboratory every day. This helps us to develop our software continuously to help tackle more capabilities as we see fit. We have also worked with our customers to help implement metrics they find important into our software.

—Tom Juel, inside sales representative, CSA Group/Orb Optronix

Analysar that could reduce LED testing time by 90 per cent

The life and light quality of LEDs has always been heavily influenced by its thermal performance. A high-tech research and development (R&D) institution, Industrial Technology Research Institute (ITRI), has developed an automatic thermal analyser that provides a measurement speed of 12,000 LEDs per hour, against traditional lab methods that offer a measurement speed of six components per hour. Touching such high thermal-resistance measurement speeds minimises the component thermal-resistance testing time of an LED to 0.3 seconds. While offering a significantly decreased testing time, InLine Compact Thermal Analyser (ICTA) aims to enhance production yield, performance and lifespan of LED devices. It uses a high-speed temperature sensitive parameter (TSP) measurement technique that looks to reduce LED testing time by more than 90 per cent.

After receiving the 2014 R&D Award, ITRI's ICTA has also bagged Frost & Sullivan's Best Practices New Product Innovation Award for 2015.



Fig. 4: ITL5000 research radiometer

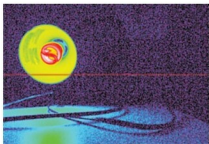


Fig. 5: High-dynamic-range mode turned off



Fig. 6: High-dynamic-range mode turned on

For very high accuracy measurements, SphereOptics introduced a new feature known as high-dynamic-range (HDR) mode for their I-Series and Y-Series ProMetric imaging photometers. For a more accurate luminance analysis, this feature can be used on a device under test to attain the lowest and highest light-illumination levels. Considering streetlights as an example of the device under test, HDR mode in I-Series and Y-Series ProMetric imaging photometers can be used to measure the illumination pattern and light directly emitted from these.

Another imaging-measurement system for quick recording of colour distributions and luminance, Konica Minolta Sensing Americas' (KMSA) and Instrument Systems' LumiCam 1300 Advanced can be used particularly for narrow-band LED colour-light measurements in today's controls and displays to meet extremely high requirements. It boasts of a unique camera concept, which uses two filters having six optical filters in total.

Lighting professionals can make use of this concept for highly accurate

measurements while analysing the characteristics of display and control elements in vehicles, flat electronic display screens, and generic lamps and luminaires used in a number of industries including aerospace. This photometer and colorimeter gives the current range of spectro-radiometric tools in the market a run for their money.

Tools to test LED drivers

Some interesting systems in the market from Chrome Instruments (Fig. 7) for testing drivers include high-precision power meters, programmable DC and AC sources and LED load simulators specifically designed for LED power drivers.

With output powers ranging from 500VA to 2000VA, the 61500 series programmable AC power sources offer three different modes, namely, AC plus DC, AC and DC output modes. These feature parameters such as measure peak and inrush current, root mean square voltage, current, power, crest factor, power factor, volt-ampere, volt-ampere reactive with programmable voltage and frequency slew rate.

On the other hand, the Chroma 66200 series are digital power meters with a wide range of power, voltage and current parameters. A user-defined benchmark provides automatic pass or fail indications, and total harmonic distortion (THD), inrush current and other energy measurements are available.

For companies into volume manufacturing of LED drivers, fixtures and arrays, it is important to verify the quality of their production swiftly. Time-to-market is an important factor to keep in mind, apart from improving product performance, validating its design and quality.

As the name suggests, ASE Test's SSL 3.0 Test System (Fig. 8) is industry's first automated test system for solid-state lighting products. It is basically used for design verification, volume manufacturing testing and performance validation. It tests DC-driven LED arrays, AC-driven LED